



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion

Executive Summary

Summary of Methodologies

1. Data Collection: Accessed SpaceX launch data via API and web scraped records from Wikipedia.
2. Data Cleaning & Preparation:
 - Cleaned and formatted the data.
 - Stored data in Db2 database and performed SQL queries.
 - Conducted exploratory data analysis.
3. Feature Engineering: Created new features and standardized the data.
4. Interactive Visualizations:
 - Mapped launch sites and success rates using Folium.
 - Built an interactive dashboard with Plotly Dash.

5. Model Building & Evaluation:

- Implemented SVM, Decision Trees, and K-Nearest Neighbors.
- Tuned hyperparameters with GridSearchCV.
- Evaluated models using test data accuracy.

Summary of Results

1. Data Insights:
 - Identified factors influencing Falcon 9 first stage landings.
 - Visualized geographical patterns and success rates.
2. Model Performance:
 - SVM and K-Nearest Neighbors: 83.33% accuracy.
 - Decision Tree: 94.44% accuracy.
3. Key Findings:
 - Launch site and payload mass impact landing success.
 - Decision Tree model is the most effective predictor.

Introduction

Project Overview and Objectives

Background

This capstone project focuses on predicting the landing success of the Falcon 9 first stage. SpaceX's competitive advantage in the launch market stems from its ability to reuse rocket stages, which significantly reduces costs. By developing an accurate predictive model for these landings, we can better estimate launch expenses and provide strategic insights for competitors in the industry.

Core Research Questions

- What variables are the primary drivers of a successful Falcon 9 landing?
- How can machine learning techniques be effectively applied to forecast landing outcomes?
- Which specific machine learning algorithm yields the highest predictive accuracy for this task?

Section 1

Methodology

Methodology

Project Overview

This project utilizes a rigorous end-to-end data science workflow—encompassing data acquisition, processing, exploratory analysis (EDA), and predictive modeling—to forecast the successful landing of the Falcon 9 first stage.

- **Data Acquisition:** Launch data, including launch sites, payload details, dates, and outcomes, was retrieved directly from the SpaceX API.
- **Data Wrangling:** The raw dataset was refined through extensive cleaning, which included managing missing values, standardizing data formats, and ensuring overall consistency. Furthermore, strategic feature engineering was employed to enhance the dataset's predictive potential.
- **Exploratory Data Analysis (EDA):**
 - **Visualization:** Matplotlib and Seaborn were used to analyze and visualize critical metrics, such as launch success rates, payload characteristics, and launch site distribution.
 - **SQL Analysis:** Targeted SQL queries were executed to extract specific insights and address key investigative questions regarding the launch records.

Methodology

Perform Interactive Visual Analytics Using Folium and Plotly Dash:

- Used Folium to create interactive maps displaying launch sites and outcomes.
- Developed a Plotly Dash application with interactive components like dropdowns and sliders to analyze launch success rates and payload ranges.

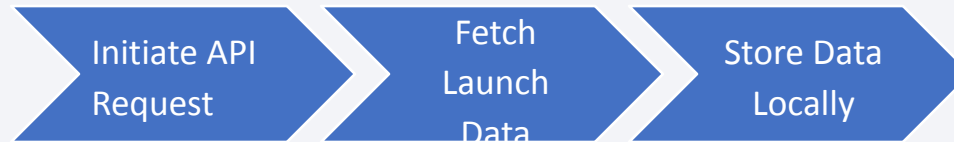
Perform Predictive Analysis Using Classification Models:

- Built and evaluated various classification models including Logistic Regression, SVM, KNN, and Decision Trees.
- Employed GridSearchCV for hyperparameter tuning.
- Evaluated models based on accuracy, and identified the best performing model for predicting landing success.

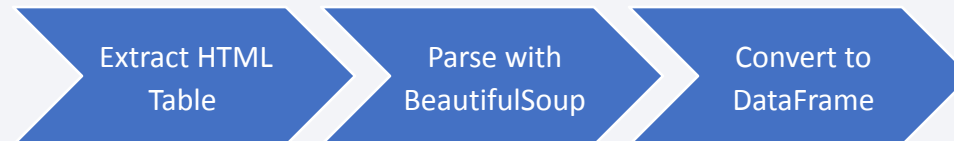
Github URL: <https://github.com/OmarKhaled06/IBM-applied-data-science-capstone-project.git>

Data Collection

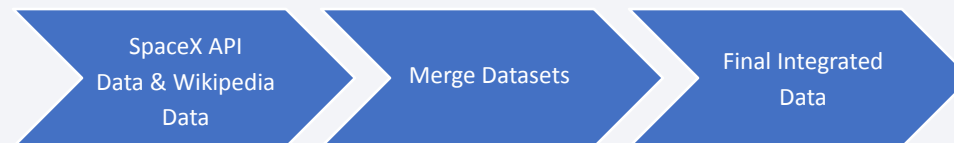
- Step 1: SpaceX API Request



- Step 2: Web Scraping Wikipedia



- Step 3: Data Integration



Data Collection – SpaceX API

Step 1: Initiate API Request

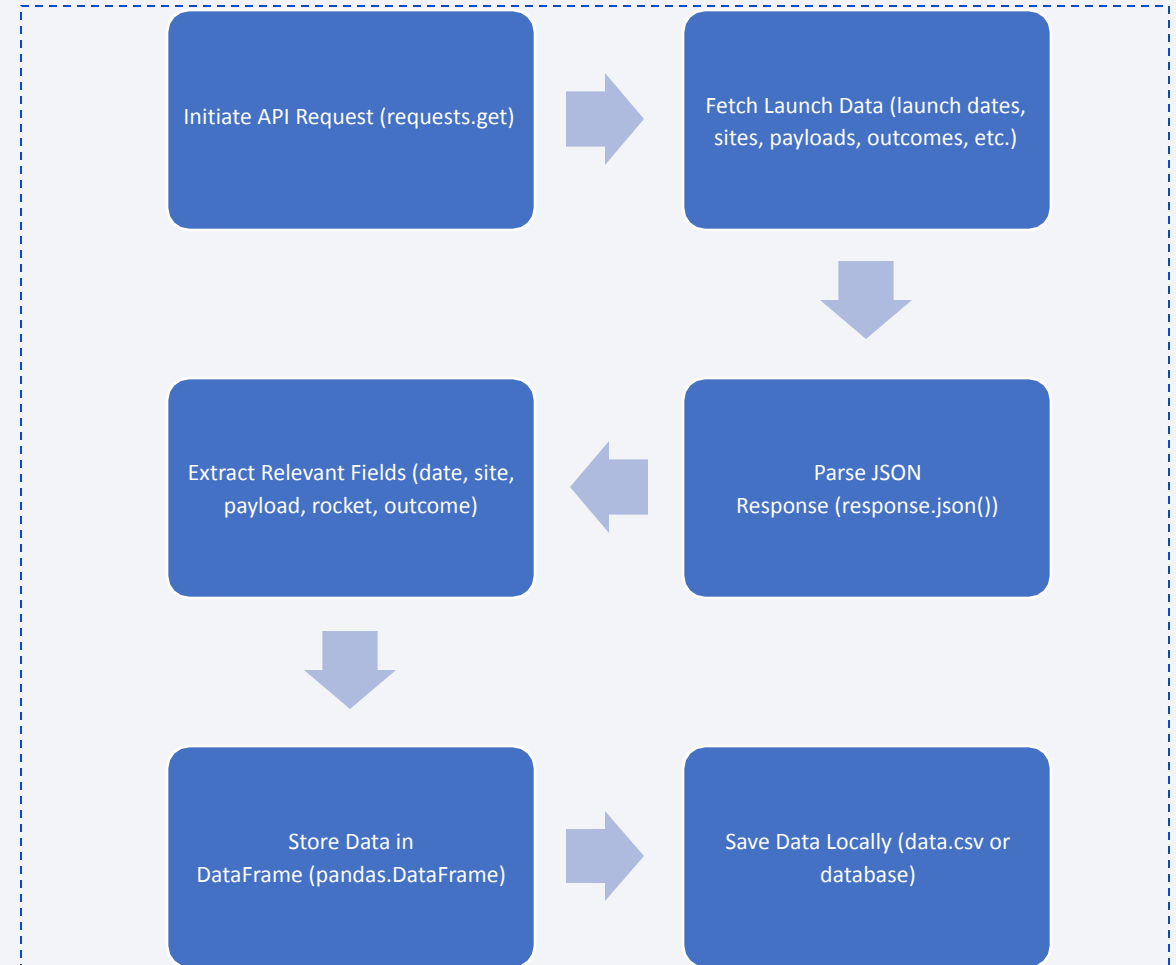
- Use Python's `requests` library to connect to the SpaceX API.
- Endpoint: `https://api.spacexdata.com/v4/launches`

Step 2: Parse API Response

- Convert API response from JSON to a Python dictionary.
- Extract relevant fields: launch date, launch site, payload mass, rocket type, outcome.

Step 3: Store Data Locally

- Save extracted data into a pandas DataFrame.
- Store the DataFrame locally for further processing.



GitHub URL:

<https://github.com/OmarKhaled06/IBM-applied-data-science-capstone-project/blob/main/1.%20jupyter-labs-spacex-data-collection-api.ipynb>

Data Collection - Scraping

Step 1: Initiate Web Scraping

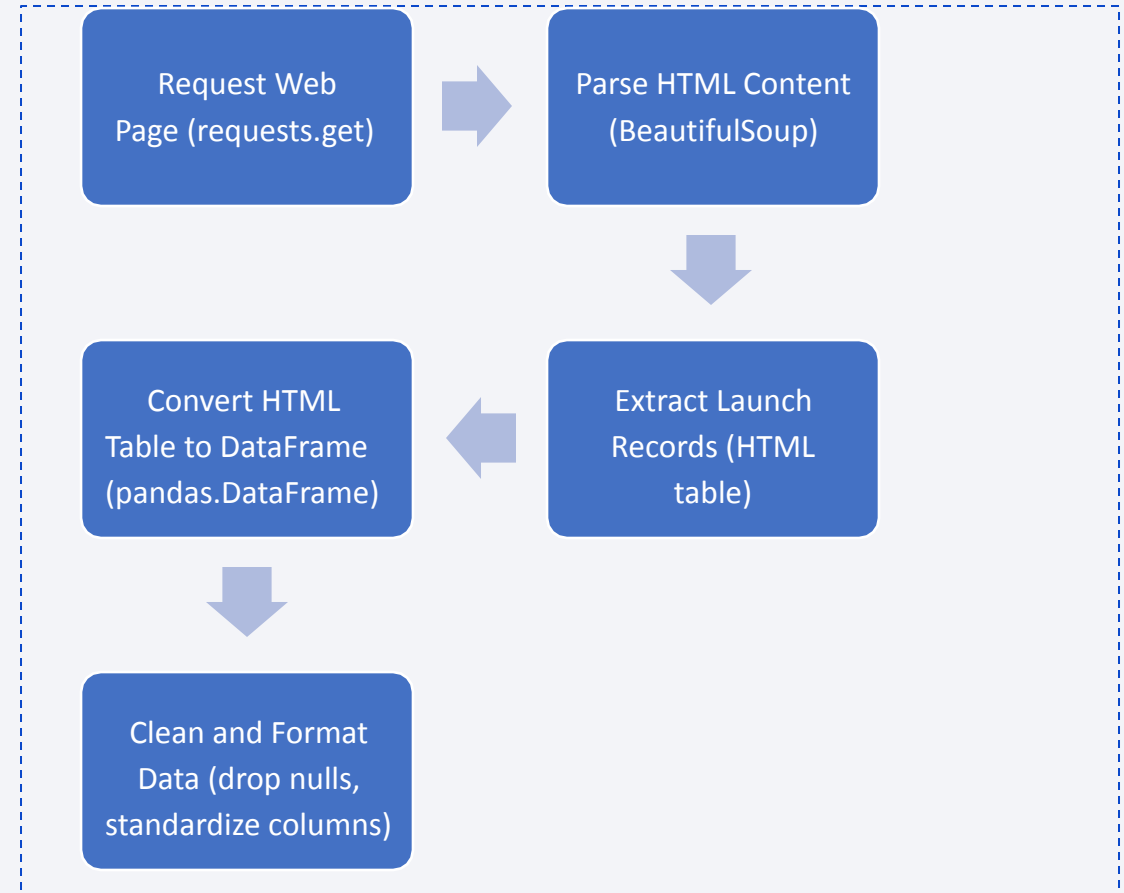
- Use Python's `requests` library to fetch the HTML content of the Wikipedia page.
- Target URL:
[`https://en.wikipedia.org/wiki/List\\_of\\_Falcon\\_9\\_and\\_Falcon\\_Heavy\\_launches`](https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches)

Step 2: Parse HTML Content

- Use `BeautifulSoup` to parse the HTML content.
- Extract the HTML table containing Falcon 9 launch records.

Step 3: Convert to DataFrame

- Convert the extracted HTML table into a pandas DataFrame.
- Clean and format the DataFrame, ensuring data consistency.



GitHub URL:

<https://github.com/OmarKhaled06/IBM-applied-data-science-capstone-project/blob/main/20jupyter-labs-webscraping.ipynb>

Data Wrangling

Data Wrangling: An Overview

Data wrangling prepares raw data for analysis through cleaning, transformation, and organization.

Phase 1: Data Cleaning

Addresses missing data by identifying gaps, imputing values, or removing records with excessive missing information.

Phase 2: Data Transformation

Refines data for consistency through:

- Formatting: Converting data types (e.g., to numerical or date-time).
- Standardization: Cleaning text (e.g., trimming whitespace, lowercasing).
- Feature Engineering: Creating new variables from existing ones.
- Scaling: Normalizing numerical features for uniformity.

Data Wrangling

Data Integration and Validation Process

Step 1: Data Integration

- Consolidate data retrieved from various channels, such as APIs and web scraping, into a unified dataset.
- Standardize column headers and data formats to ensure uniformity throughout the merged collection.

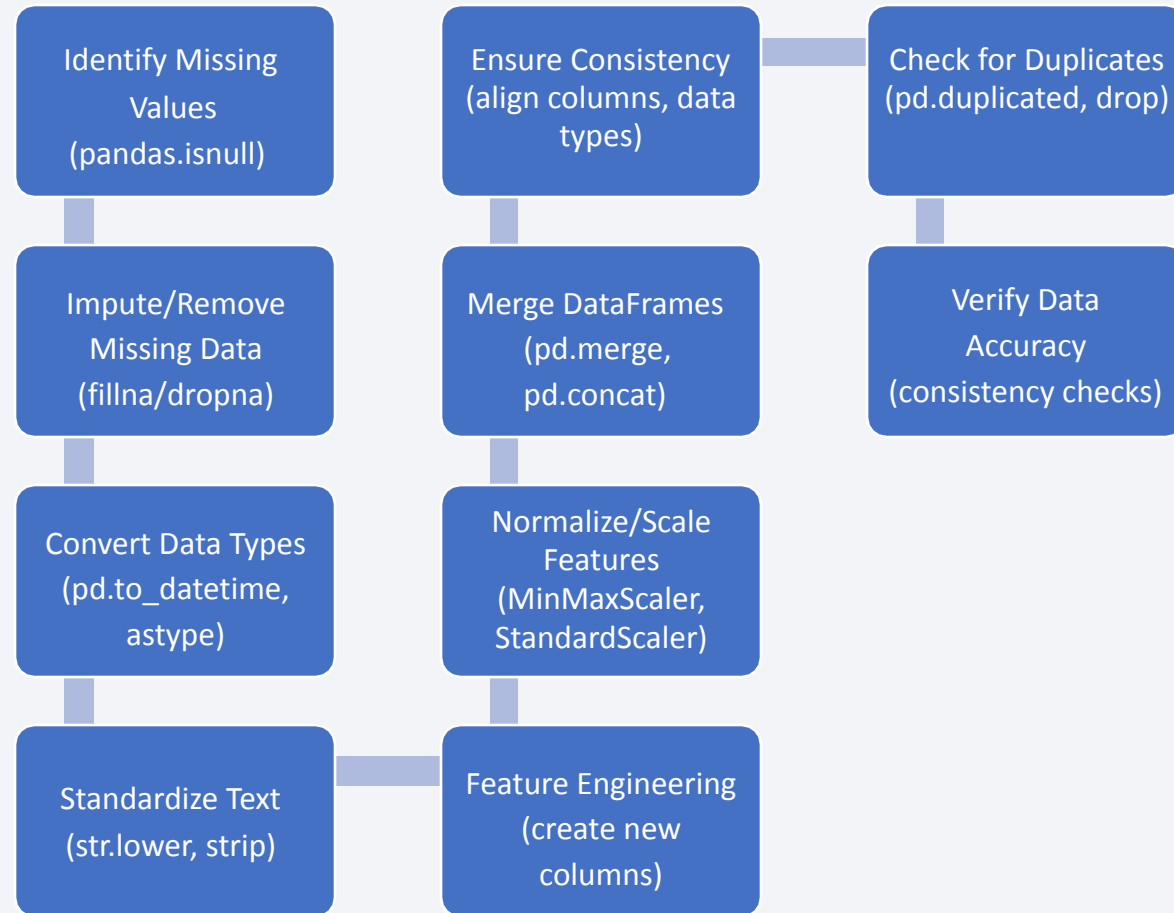
Step 2: Data Validation

- Identify and eliminate duplicate records to maintain data integrity.
- Perform quality assurance checks to confirm the accuracy and consistency of all entries.

GitHub URL:

<https://github.com/OmarKhaled06/IBM-applied-data-science-capstone-project/blob/main/2.%20jupyter-labs-webscraping.ipynb>

Data Wrangling Flowchart



EDA with Data Visualization

Overview:

Exploratory Data Analysis (EDA) involves visually exploring and summarizing the main characteristics of a dataset. The goal is to understand the data's distribution, identify patterns, and uncover relationships between variables.

Charts Plotted:

1. Histograms:

- **Purpose:** Used to visualize the distribution of numerical variables such as launch success rates, payload mass, and flight number.
- Why:** Helps in understanding the spread and central tendency of the data, identifying outliers, and assessing data skewness.

2. Bar Charts:

- **Purpose:** Used to compare categorical variables such as launch outcomes (success/failure) across different categories like launch sites or rocket types.
- Why: Provides a clear comparison of frequencies or proportions within categorical data, highlighting patterns or trends.

3. Line Charts:

- Purpose:** Used to track trends over time, such as the success rate of Falcon 9 launches across different years.
- Why:** Reveals temporal patterns and helps in understanding performance trends or changes over specific periods.

EDA with Data Visualization

4. Scatter Plots:

- Purpose:** Used to explore relationships between two numerical variables, such as payload mass vs. launch success.
- Why:** Identifies correlations or dependencies between variables, visualizing how one variable changes concerning another.

5. Heatmaps:

- Purpose:** Used to visualize correlation matrices between multiple numerical variables.
- Why:** Helps in identifying strong correlations (positive or negative) between variables, aiding feature selection or understanding multicollinearity.

6. Box Plots:

- Purpose:** Used to display the distribution of numerical data through their quartiles.
- Why:** Visualizes the spread and skewness of data, highlighting outliers and comparing distributions across different categories.

Github

URL:<https://github.com/OmarKhaled06/IBM-applied-data-science-capstone-project/blob/main/5.%20jupyter-labs-eda-dataviz.ipynb>

EDA with SQL

Aggregate Queries:

- Calculated total number of launches.
- Counted successful and failed launches.
- Calculated success rates by launch site and rocket type.

Join Queries:

- Joined tables to link launch records with additional data (e.g., rocket details).
- Combined datasets for comprehensive analysis.

Filtering Queries:

- Filtered data to focus on specific launch outcomes (success/failure).
- Applied conditions to extract launches based on criteria like launch date or rocket configuration.

Sorting Queries:

- Sorted data to identify trends or outliers.
- Ordered launches by date or success rate for analysis.

Subqueries:

- Nested queries to calculate derived metrics (e.g., average payload mass per launch site).
- Subqueries used to perform detailed analysis within larger datasets.

GitHub URL: [4. jupyter-labs-eda-sql-coursera_sqlite.ipynb](#)

Build an Interactive Map with Folium

Map Objects Created

Markers:

- Placed markers to indicate launch sites on the map.
- Each marker represents a specific geographical location where SpaceX launches have occurred.

Circles:

- Added circles around launch sites to visually represent proximity zones.
- Circles help visualize the areas around launch sites that might influence operational decisions.

Lines:

- Drew lines to connect launch sites with their proximities or other relevant locations.
- Lines provide spatial context and connections between different points of interest related to launches.

Github URL:

https://github.com/OmarKhaled06/IBM-applied-data-science-capstone-project/blob/main/4.%20jupyter-labs-eda-sql-coursera_sqlite.ipynb

Reasons for Adding Objects

Markers:

- To pinpoint exact launch locations for spatial reference.
- Helps users identify where SpaceX has conducted launches geographically.

Circles:

- Illustrates the potential impact zones around launch sites.
- Provides a visual representation of safety perimeters or operational boundaries.

Lines:

- Shows connections or relationships between launch sites and relevant features.
- Enhances understanding of spatial relationships and dependencies.

Build a Dashboard with Plotly Dash

Plots/Graphs Added

Success Pie Chart:

- Displays the distribution of successful and failed launches.
- Helps visualize the overall success rate and performance trends.

Success-Payload Scatter Plot:

- Shows the relationship between payload mass and launch success.
- Allows users to explore how payload mass influences mission outcomes.

Github URL:

https://github.com/OmarKhaled06/IBM-applied-data-science-capstone-project/blob/main/6.%20lab_jupyter_launch_site_location.ipynb

Interactions Added

Launch Site Dropdown:

- Enables users to select specific launch sites for analysis.
- Facilitates filtering and focused exploration based on geographical locations.

Range Slider for Payload:

- Allows users to adjust payload mass ranges dynamically.
- Offers flexibility in examining launch success concerning payload mass variations.

Reasons for Adding Plots and Interactions

Success Pie Chart:

- Provides a quick overview of mission success rates.
- Essential for stakeholders to understand overall performance metrics at a glance.

Success-Payload Scatter Plot:

- Helps identify correlations between payload characteristics and launch outcomes.
- Supports decision-making processes related to payload planning and operational strategies.

Launch Site Dropdown:

- Enhances user experience by focusing analysis on specific launch locations.
- Allows for regional insights and comparisons across different launch sites.

Range Slider for Payload:

- Offers interactive exploration of how payload mass affects mission success.
- Enables detailed analysis and insights into payload-related performance factors.

Predictive Analysis (Classification)

1- Data Preprocessing

- **Feature Standardization:** Applied scaling techniques to ensure all variables contribute equitably to the model.
- **Data Partitioning:** Divided the dataset into training and testing sets to facilitate robust model validation.

2- Model Selection

- **Algorithm Exploration:** Evaluated several classification approaches, including Support Vector Machines (SVM), Decision Trees, and K-Nearest Neighbors (KNN).
- **Suitability:** Selected these algorithms specifically for their effectiveness in handling binary classification objectives.

3- Hyperparameter Tuning

- **Optimization Strategy:** Employed **GridSearchCV** to conduct a systematic search for optimal hyperparameter configurations.
- **Key Parameters Adjusted:** Refined critical parameters, specifically **C** for SVM, **max_depth** for Decision Trees, and **n_neighbors** for KNN.
- Github url:
https://github.com/OmarKhaled06/IBM-applied-data-science-capstone-project/blob/main/7.%20SpaceX_Machine_Learning_Prediction_Part_5.jupyterlite.ipynb

4. Model Evaluation:

- Evaluated models using cross-validation techniques to ensure robustness and generalizability.
- Utilized metrics like accuracy, precision, recall, and F1-score to assess model performance.

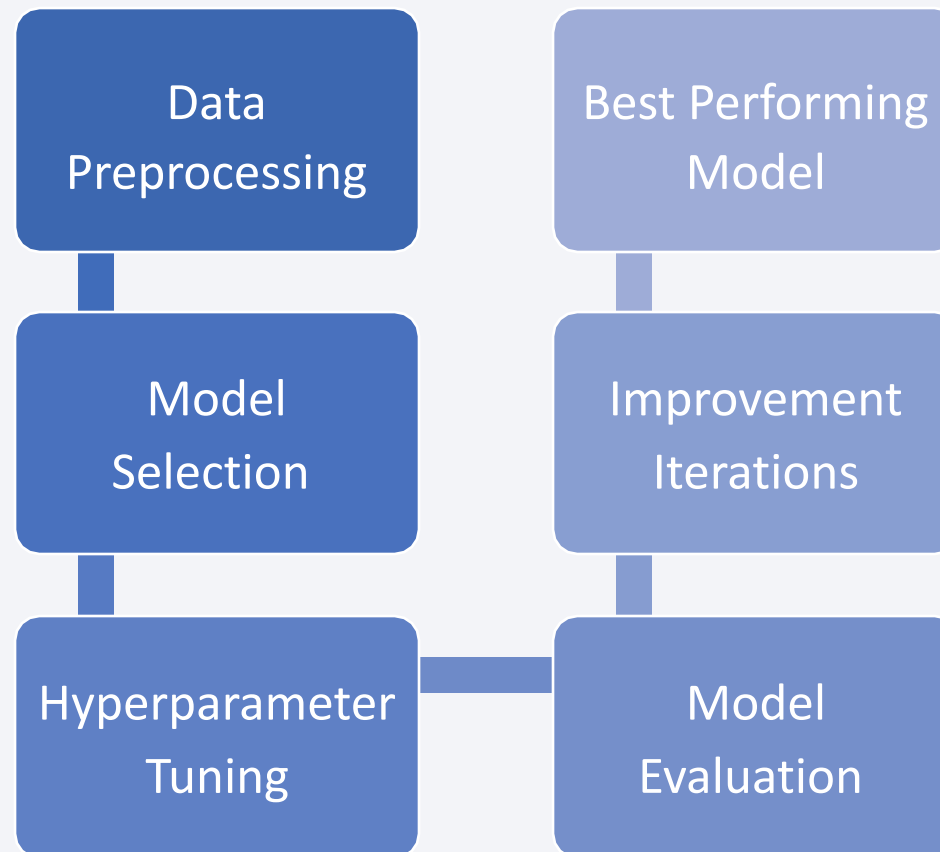
5. Improvement Iterations:

- Iteratively adjusted models based on insights from validation results.
- Fine-tuned hyperparameters to maximize predictive accuracy and reliability.

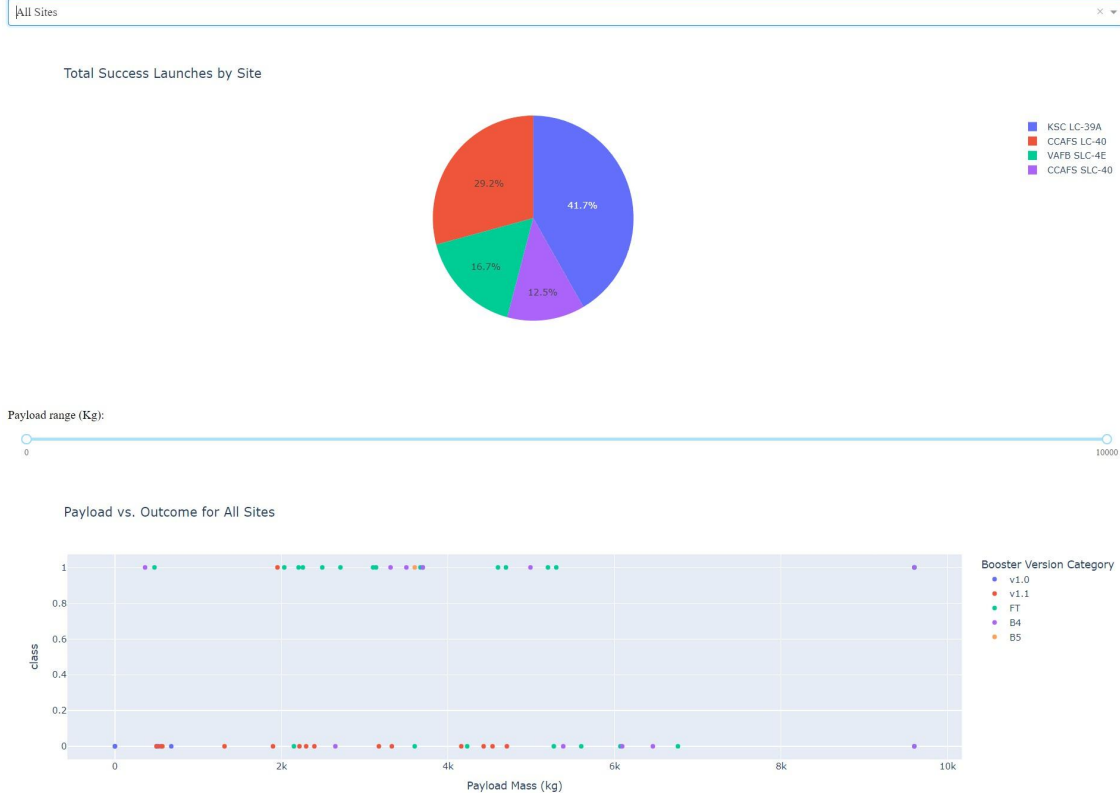
6. Selection of Best Performing Model:

- Identified the model with the highest accuracy on the test set as the best performer.
- Considered both training and test set performance to avoid overfitting and ensure real-world applicability.

Predictive Analysis (Flowchart)



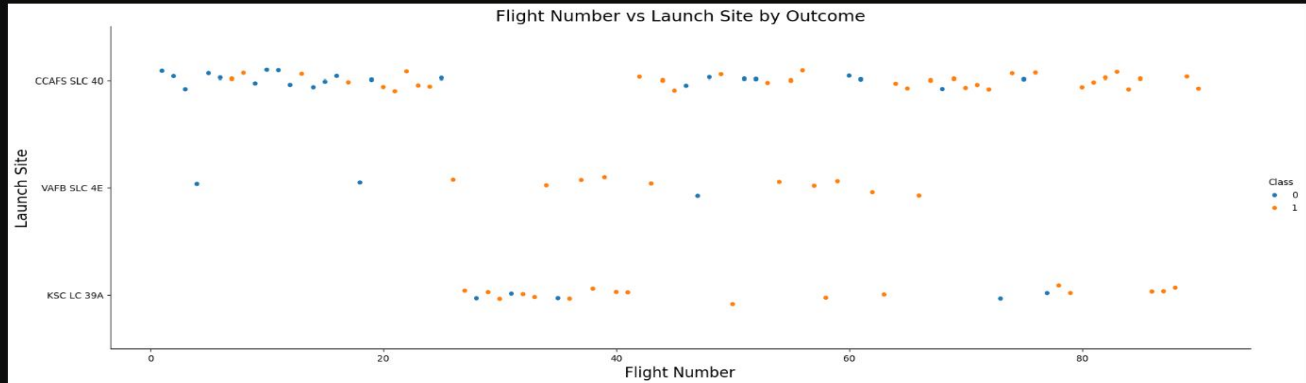
SpaceX Launch Records Dashboard



TASK 1: Visualize the relationship between Flight Number and Launch Site

Use the function `catplot` to plot `FlightNumber` vs `LaunchSite`, set the parameter `x` parameter to `FlightNumber`, set the `y` to `Launch Site` and set the parameter `hue` to `'class'`

```
[10]: # Plot a scatter point chart with x axis to be Flight Number and y axis to be the Launch site, and hue to be the class value
sns.catplot(x="FlightNumber", y="LaunchSite", hue="Class", data=df, aspect=3, height=6, kind="strip")
plt.xlabel("Flight Number", fontsize=16)
plt.ylabel("Launch Site", fontsize=16)
plt.title("Flight Number vs Launch Site by Outcome", fontsize=18)
plt.show()
```



TASK 12

Find the method performs best:

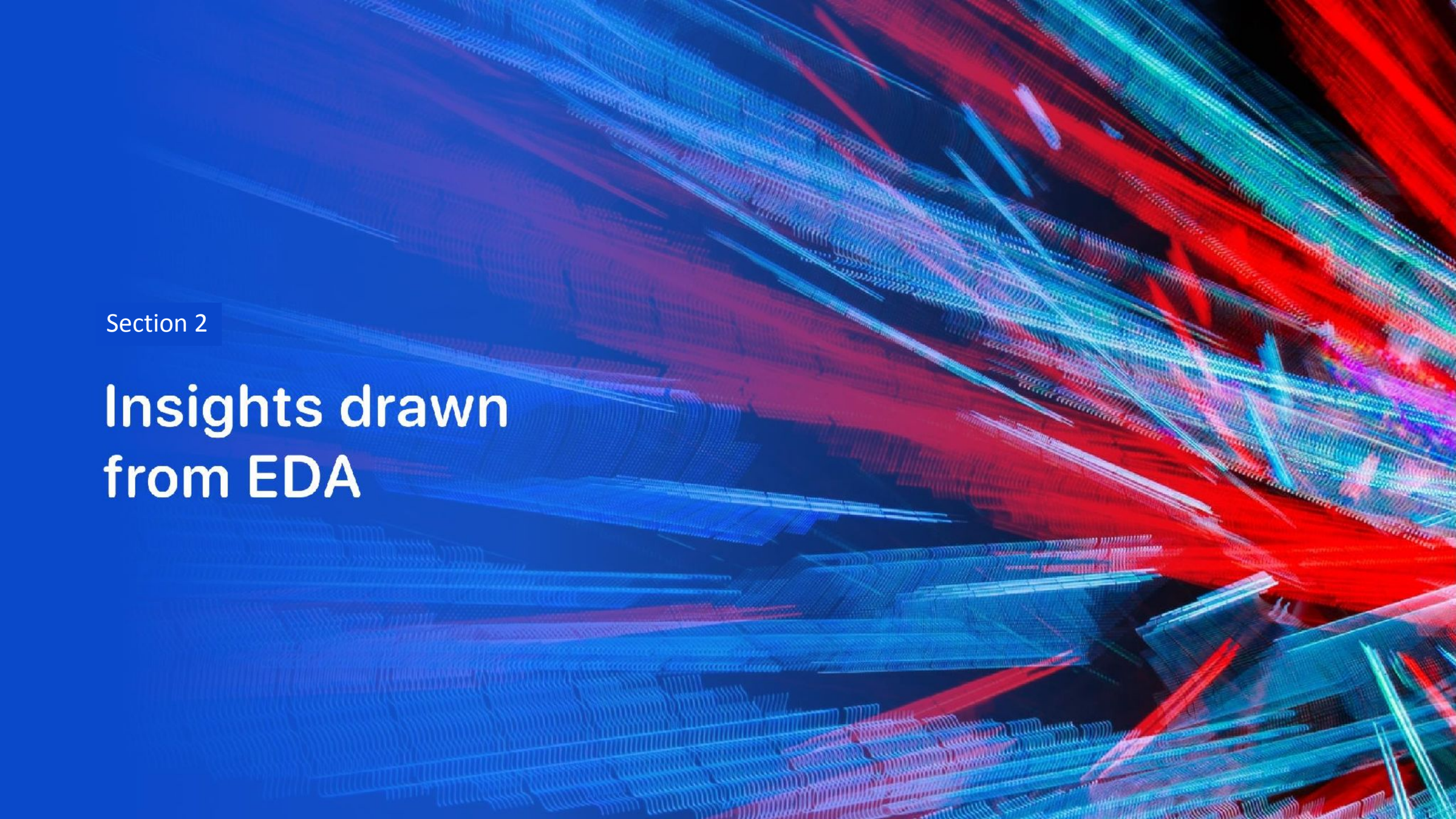
Based on the results I got:

1. **Logistic Regression:** Accuracy on test data: 0.8333
2. **Support Vector Machine (SVM):** Accuracy on test data: 0.8333
3. **Decision Tree:** Accuracy on test data: 0.9444
4. **K Nearest Neighbors (KNN):** Accuracy on test data: 0.8333

It seems there was a significant increase in accuracy for the Decision Tree model, which shows the highest accuracy of 0.9444 on the test set. Therefore, in this comparison, the Decision Tree model performs the best among the classifiers you've trained.

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

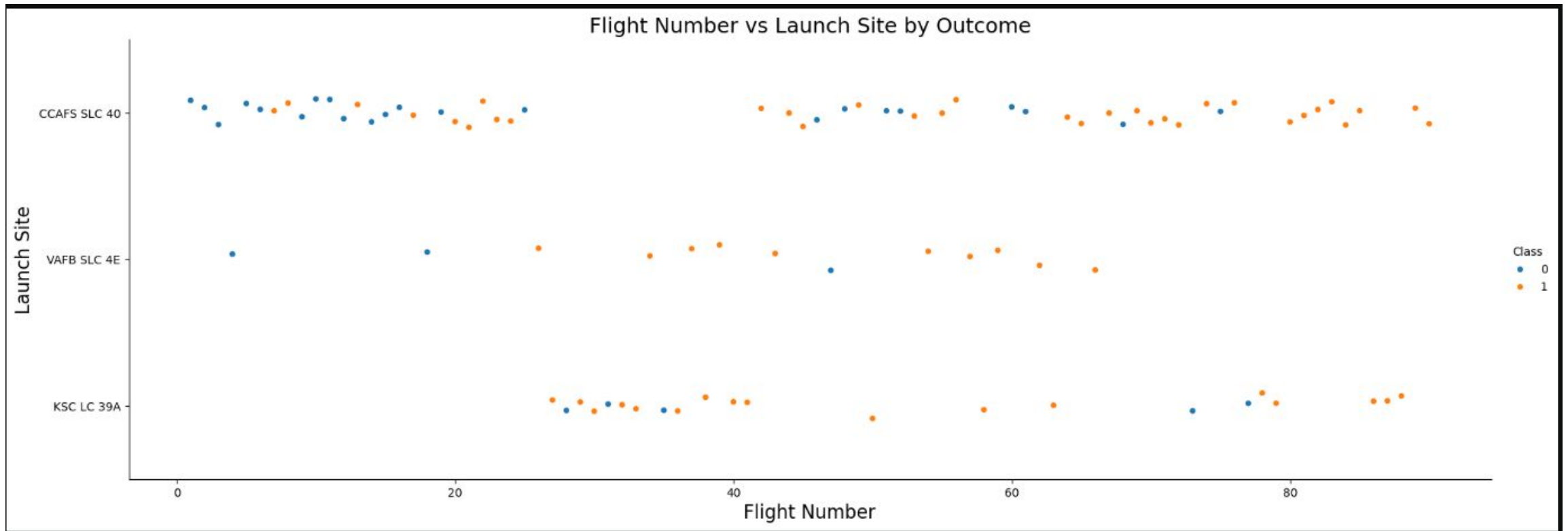
The background of the slide is an abstract composition. It features a solid blue field on the left side, which transitions into a complex pattern of diagonal streaks and lines in shades of blue, red, and cyan on the right. These streaks have a textured, almost woven appearance, suggesting a digital or data-driven theme. The overall effect is dynamic and modern.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

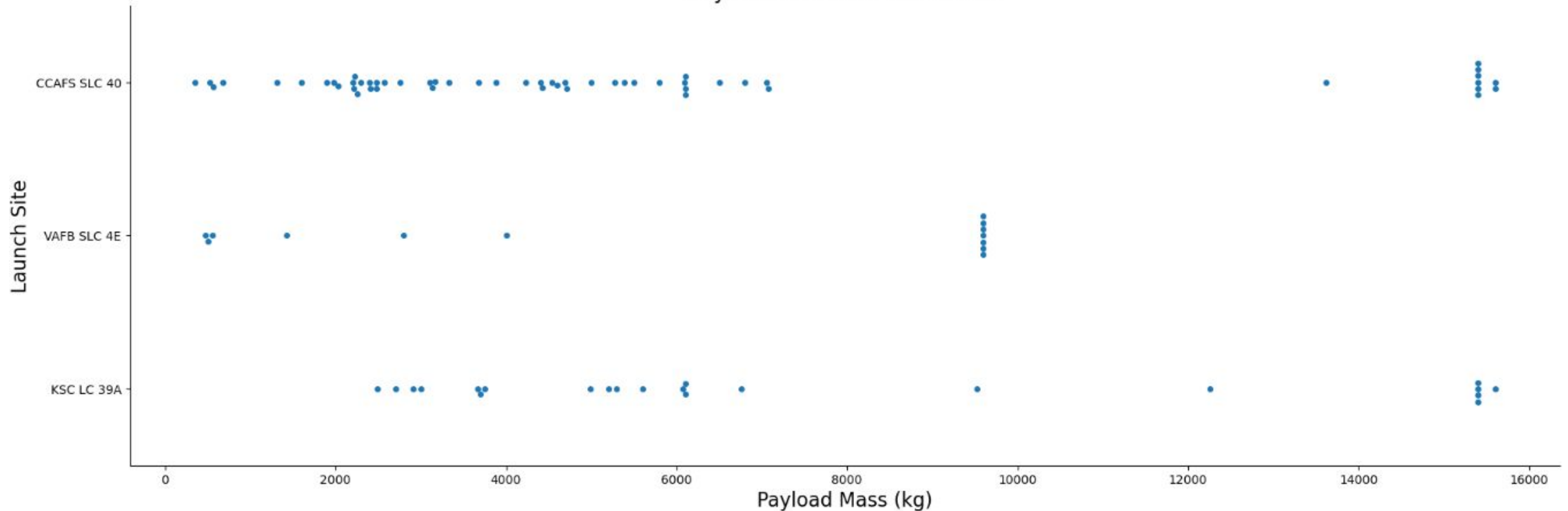
- **Mixed Outcomes at Major Launch Sites:** Both CCAFS SLC 40 and KSC LC 39A have a mix of successful (orange) and unsuccessful (blue) landings, indicating that factors other than the launch site itself may influence the landing success.
- **Consistent Activity Across Flight Numbers:** Launches are spread across a wide range of flight numbers at all sites, suggesting consistent activity over time without a clear trend of increasing or decreasing landing success.



Payload vs. Launch Site

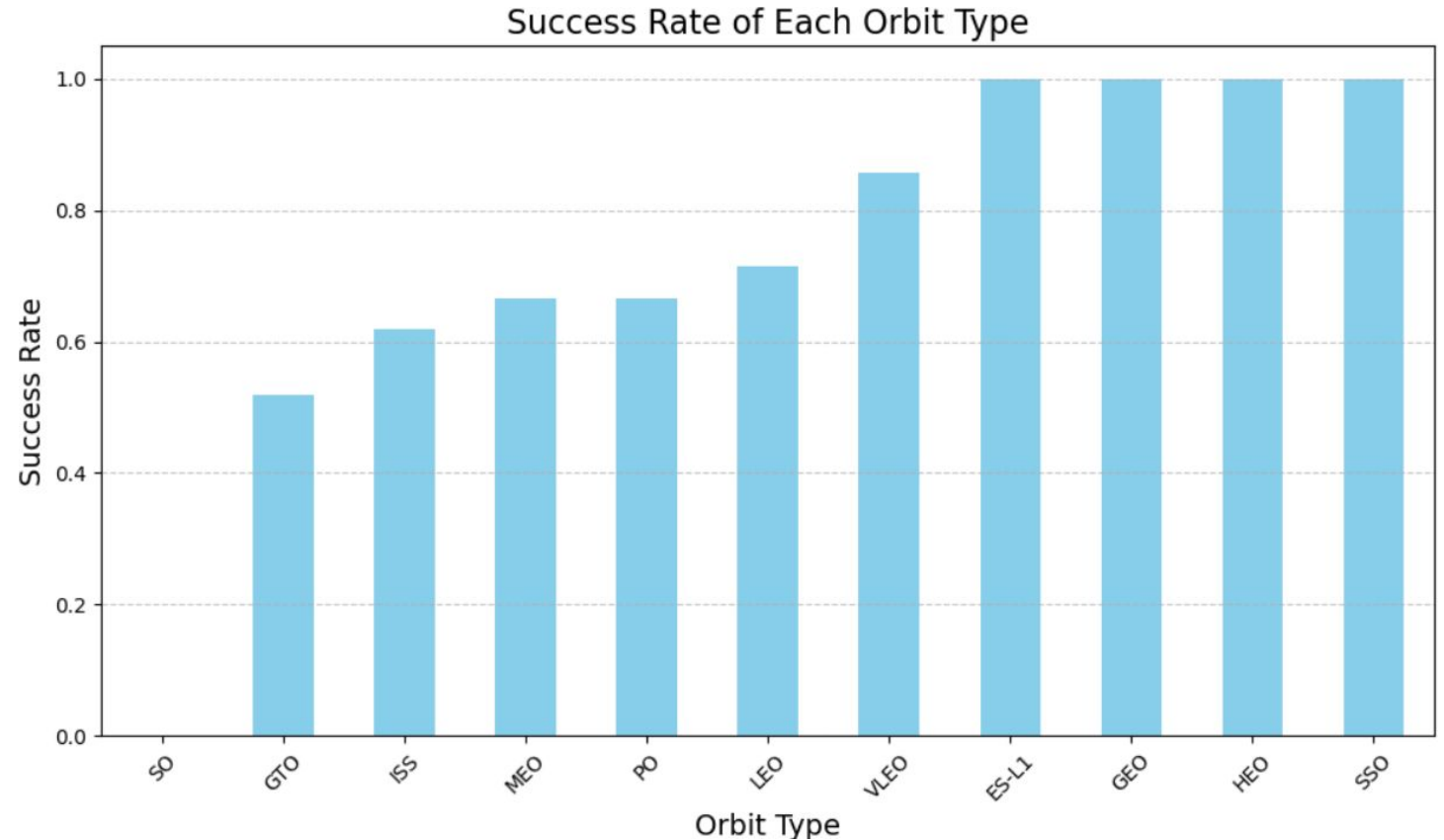
- **Payload Distribution:** Most launches from the CCAFS SLC 40 site handle payloads below 10,000 kg, while the VAFB SLC 4E and KSC LC 39A sites have a wider range of payload masses, indicating varied mission profiles.
- **High-Capacity Launches:** The KSC LC 39A site is frequently used for launching heavier payloads, with multiple launches carrying over 15,000 kg, suggesting its suitability for high-capacity missions.

Payload Mass vs Launch Site



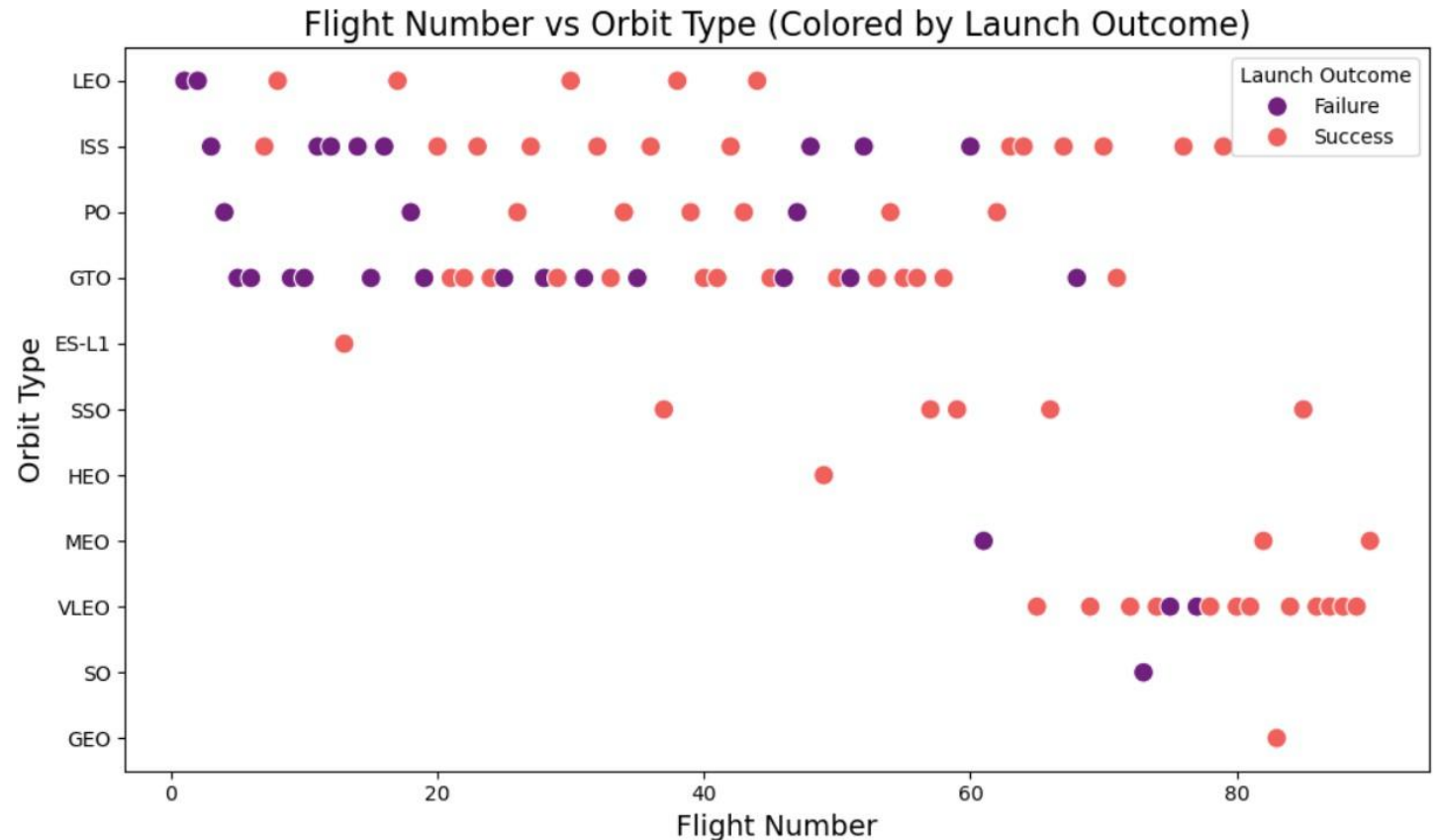
Success Rate vs. Orbit Type

- **High Success Rates:** Missions to VLEO, ES-L1, GEO, HEO, and SSO orbits have achieved a perfect success rate, indicating these orbits are highly reliable for successful first stage landings.
- **Lower Success Rate for GTO:** The GTO orbit type shows a significantly lower success rate compared to other orbit types, suggesting that missions to this orbit may involve greater challenges or complexities.



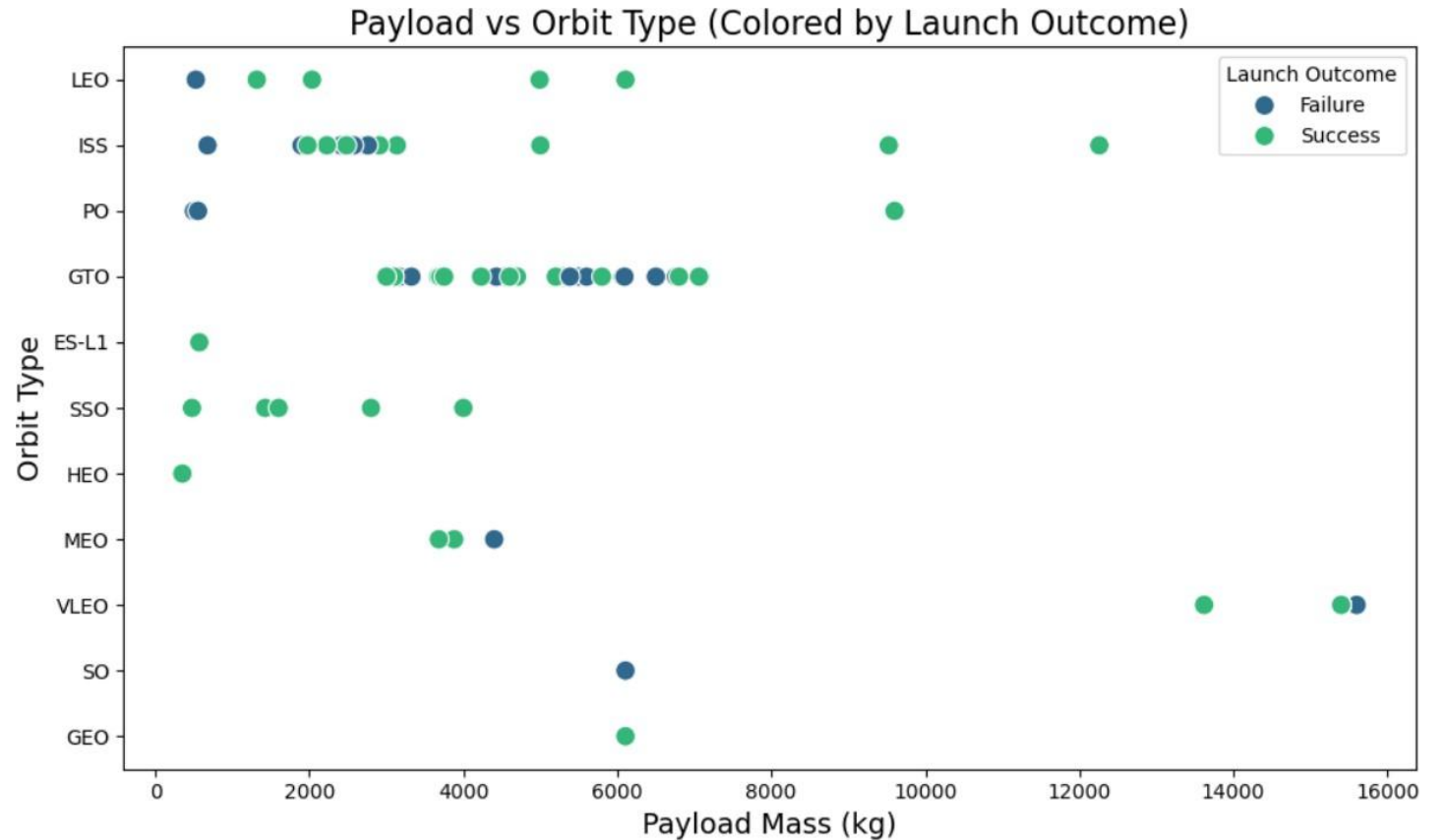
Flight Number vs. Orbit Type

- **Increased Success Over Time:** The success rate of Falcon 9 launches improves significantly with higher flight numbers, indicating that experience and iterative improvements contribute to better outcomes.
- **Orbit-Specific Performance:** Early flights to GTO and ISS orbits had mixed outcomes, but recent missions to these orbits show a higher success rate, reflecting advancements in mission planning and execution.



Payload vs. Orbit Type

- Successful landings are more frequent across all orbit types, especially for payloads less than 6000 kg.
- Higher payload masses (above 10,000 kg) show a mix of successes and failures, indicating increased difficulty with heavier payloads.



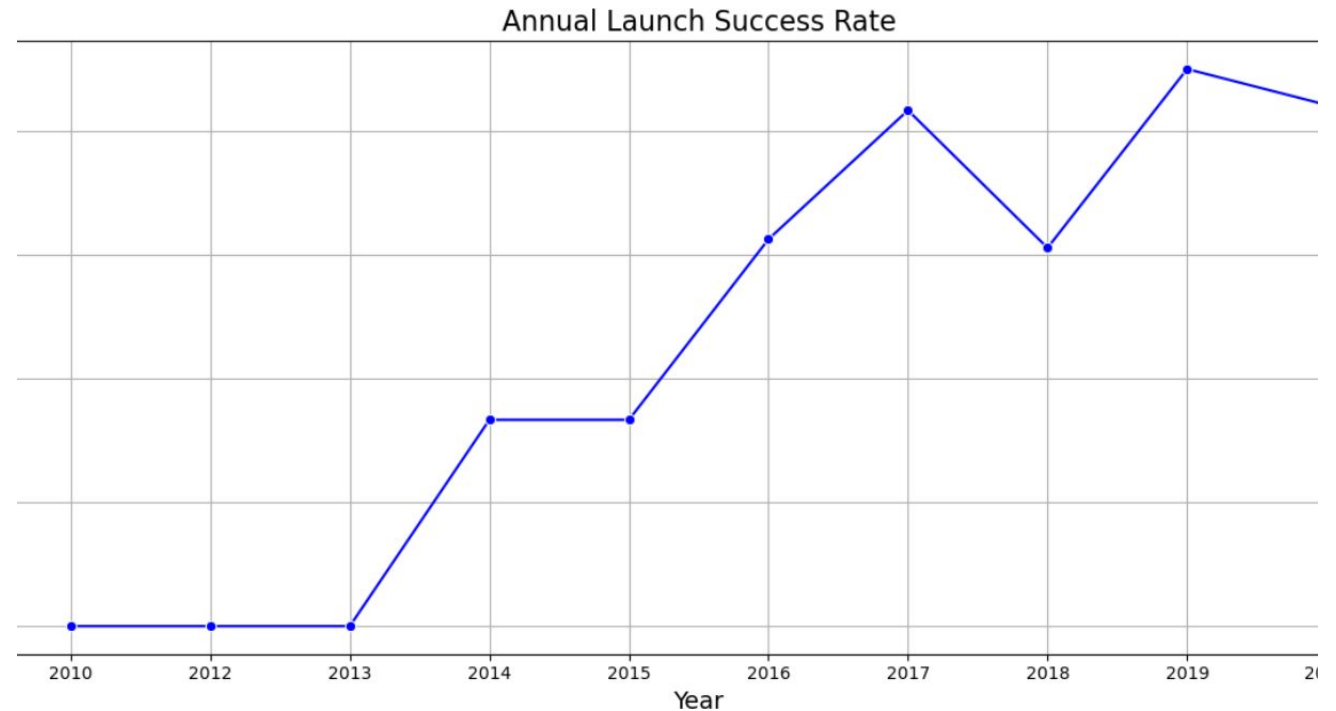
Launch Success Yearly Trend

Significant Growth

The annual launch success rate has shown a significant improvement from 2013 onwards, reaching over **80% by 2020**.

Long-Term Reliability

Despite a temporary dip in 2018, the overall trend indicates **increasing reliability** and success in Falcon 9 launches over the years.



All Launch Site Names

Task 1

Display the names of the unique launch sites in the space mission

```
[21]: %sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db
```

Done.

```
[21]: Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

Launch Site Names Begin with 'CCA'

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
[26]: %sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db
```

Done.

```
[26]:
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
[30]: %sql SELECT SUM("PAYLOAD_MASS_KG_") FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[30]: SUM(PAYLOAD_MASS_KG_)
```

```
45596
```

Average Payload Mass by F9 v1.1

Task 4

Display average payload mass carried by booster version F9 v1.1

```
[34]: %sql SELECT AVG("PAYLOAD_MASS_KG_") FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

Done.

```
[34]: AVG(PAYLOAD_MASS_KG_)
```

```
2928.4
```

First Successful Ground Landing Date

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
[36]: %sql SELECT MIN("Date") FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';  
* sqlite:///my_data1.db  
Done.
```

```
[36]: MIN(Date)  
2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
[38]: %sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS_KG_" > 4000 AND "PAYLOAD_MASS_KG_" < 6000

* sqlite:///my_data1.db
Done.
```

```
[38]: Booster_Version
```

```
F9 FT B1022
```

```
F9 FT B1026
```

```
F9 FT B1021.2
```

```
F9 FT B1031.2
```


Total Number of Successful and Failure Mission Outcomes

Task 7

List the total number of successful and failure mission outcomes

```
[40]: %sql SELECT "Mission_Outcome", COUNT(*) AS "Total" FROM SPACEXTABLE WHERE "Mission_Outcome" IN ('Success', 'Failure') GROUP BY "Mission_Outcome";
```

```
* sqlite:///my_data1.db
```

Done.

```
[40]:
```

Mission_Outcome	Total
Success	98

Boosters Carried Maximum Payload

Task 8

List the names of the booster_versions which have carried the maximum payload mass. Use a subquery

```
[42]: %sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM SPACEXTABLE);
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[42]: Booster_Version
```

```
F9 B5 B1048.4
```

```
F9 B5 B1049.4
```

```
F9 B5 B1051.3
```

```
F9 B5 B1056.4
```

```
F9 B5 B1048.5
```

```
F9 B5 B1051.4
```

```
F9 B5 B1049.5
```

```
F9 B5 B1060.2
```

```
F9 B5 B1058.3
```

```
F9 B5 B1051.6
```

```
F9 B5 B1060.3
```

```
F9 B5 B1049.7
```

2015 Launch Records

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
[69]: %%sql
SELECT
  CASE
    WHEN substr("Date", 6, 2) = '01' THEN 'January'
    WHEN substr("Date", 6, 2) = '02' THEN 'February'
    WHEN substr("Date", 6, 2) = '03' THEN 'March'
    WHEN substr("Date", 6, 2) = '04' THEN 'April'
    WHEN substr("Date", 6, 2) = '05' THEN 'May'
    WHEN substr("Date", 6, 2) = '06' THEN 'June'
    WHEN substr("Date", 6, 2) = '07' THEN 'July'
    WHEN substr("Date", 6, 2) = '08' THEN 'August'
    WHEN substr("Date", 6, 2) = '09' THEN 'September'
    WHEN substr("Date", 6, 2) = '10' THEN 'October'
    WHEN substr("Date", 6, 2) = '11' THEN 'November'
    WHEN substr("Date", 6, 2) = '12' THEN 'December'
    ELSE 'Unknown'
  END AS "Month_Name",
  "Mission_Outcome",
  "Booster_Version",
  "Launch_Site"
FROM
  SPACEXTABLE
WHERE
  substr("Date", 0, 5) = '2015';

* sqlite:///my_data1.db
Done.
```

```
[69]:
```

Month_Name	Mission_Outcome	Booster_Version	Launch_Site
January	Success	F9 v1.1 B1012	CCAFS LC-40
February	Success	F9 v1.1 B1013	CCAFS LC-40
March	Success	F9 v1.1 B1014	CCAFS LC-40
April	Success	F9 v1.1 B1015	CCAFS LC-40
April	Success	F9 v1.1 B1016	CCAFS LC-40
June	Failure (in flight)	F9 v1.1 B1018	CCAFS LC-40
December	Success	F9 FT B1019	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
[81]: %%sql
SELECT
    "Landing_Outcome",
    COUNT(*) AS "Count"
FROM
    SPACEXTABLE
WHERE
    "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY
    "Landing_Outcome"
ORDER BY
    COUNT(*) DESC;
```

```
* sqlite:///my_data1.db
```

Done.

```
[81]:
```

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis



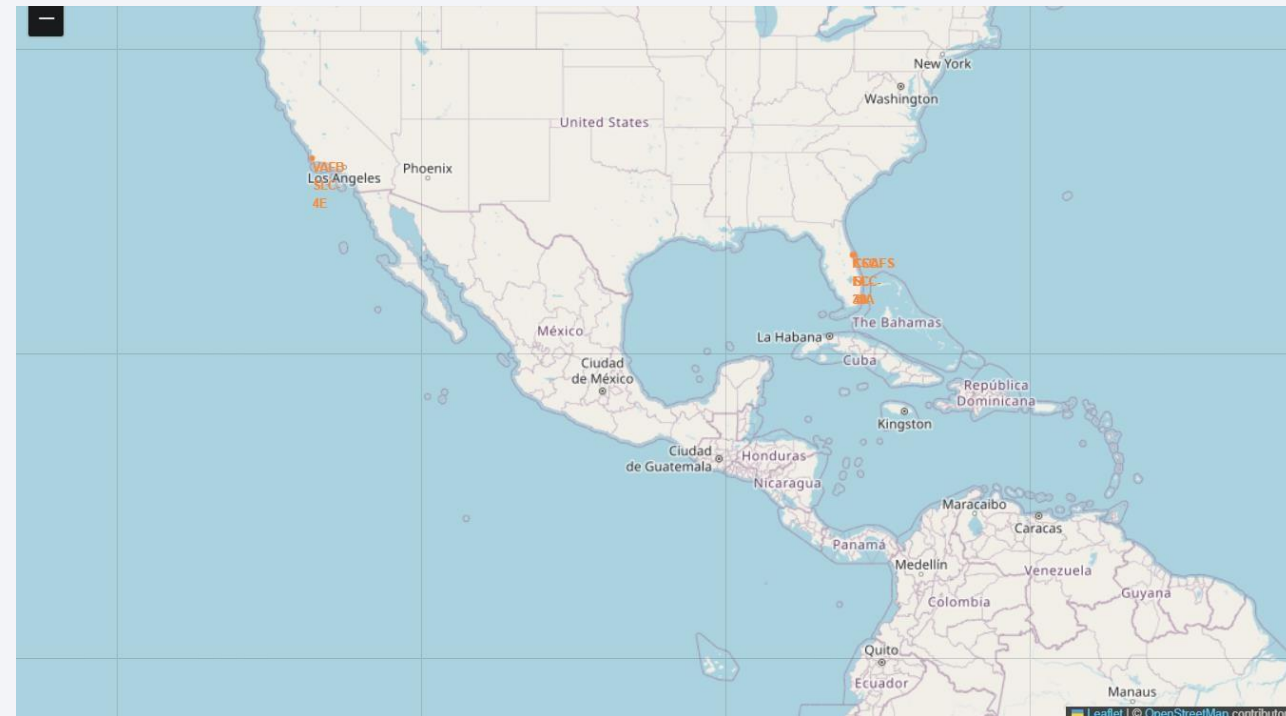
Task 1: Mark all launch sites on a map

1. Are all launch sites in proximity to the Equator line?

- No, not all launch sites are in close proximity to the Equator.
- The launch site at Vandenberg Air Force Base (VAFB SLC-4E) is located at a latitude of 34.63, which is further from the Equator compared to the other sites in Florida.

2. Are all launch sites in very close proximity to the coast?

- Yes, all launch sites are in close proximity to the coast.
- The Cape Canaveral sites (CCAFS LC-40 and CCAFS SLC-40) and Kennedy Space Center (KSC LC-39A) are near the coast in Florida.
- Vandenberg Air Force Base (VAFB SLC-4E) is also near the coast in California.



Task 2: Mark the success/failed launches for each site on the map

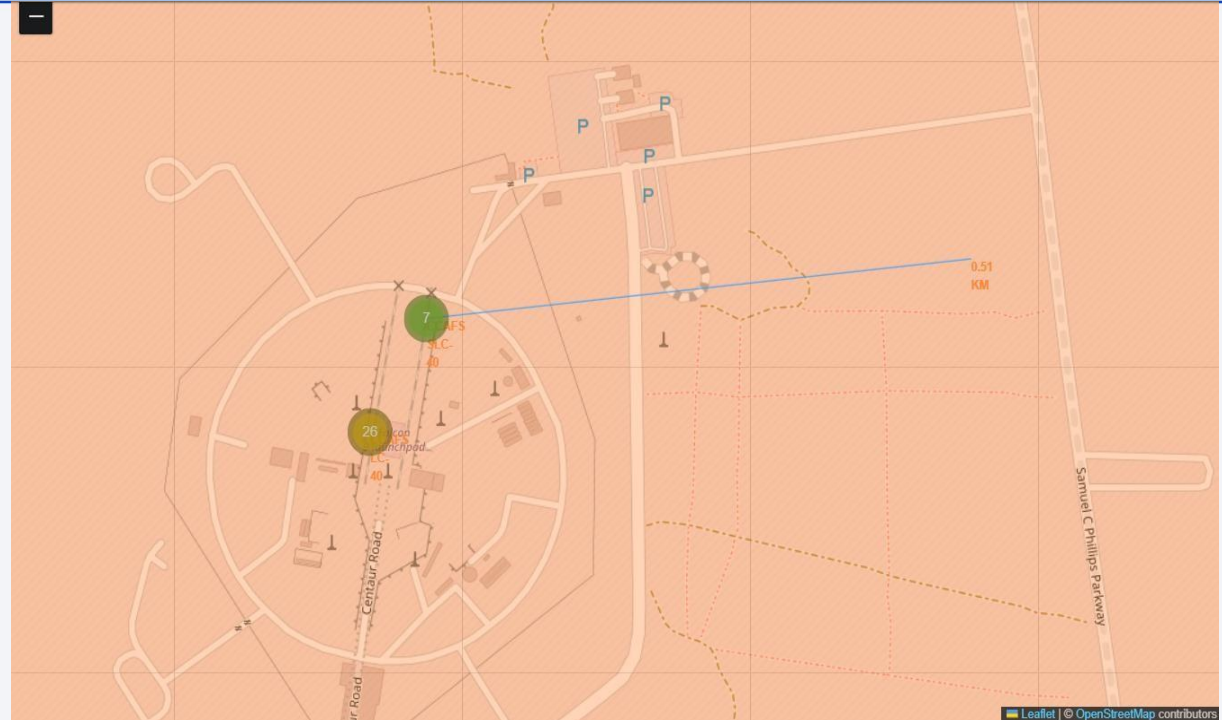
By utilizing clustered markers, this visualization significantly improves the exploration and analysis of SpaceX launch data. This method simplifies the management of high-density data points, revealing patterns that might otherwise be obscured. Key insights into the distribution and characteristics of these launches can be extracted by analyzing marker colors and popup details.

For instance, the screenshot displays a breakdown for the CCAFS LC-40 launch site, showing 19 red markers and 7 green markers. This color-coded approach enables rapid assessment of launch outcomes—where green likely signifies success and red indicates failure—providing immediate visual confirmation of performance trends at individual sites.



Task 3: Calculate the distances between a launch site to its proximities

This plot provides a visual representation of the distance between the CCAFS SLC-40 launch site and the closest coastline. The calculated distance is approximately 0.51 kilometers, as indicated by the marker. The added PolyLine clearly shows the straight-line distance, highlighting the proximity of the launch site to the coast. This close proximity to the coastline is typical for launch sites to facilitate over-water flight paths and safe recovery operations, ensuring minimal risk to populated areas.



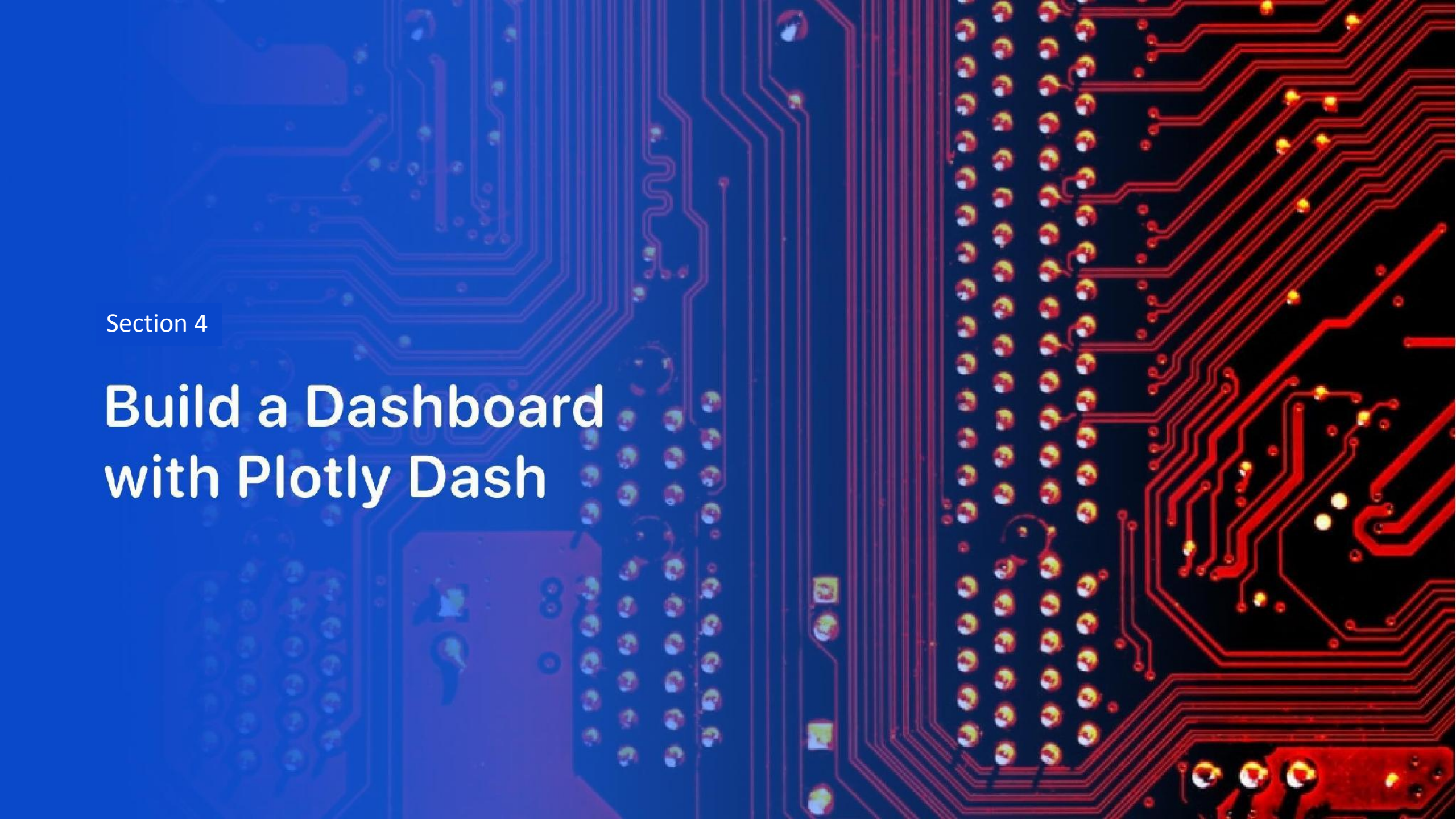
```
[43]: coastline_lat = 28.56367
      coastline_lon = -80.57163

      # Example launch site coordinates (replace with actual launch site coordinates)
      launch_site_lat = launch_sites_df.loc[launch_sites_df['Launch Site'] == 'CCAFS SLC-40', 'Lat'].values[0]
      launch_site_lon = launch_sites_df.loc[launch_sites_df['Launch Site'] == 'CCAFS SLC-40', 'Long'].values[0]

      # Calculate distance using the calculate_distance function
      distance_coastline = calculate_distance(launch_site_lat, launch_site_lon, coastline_lat, coastline_lon)

      print(f"Distance from launch site to closest coastline: {distance_coastline} km")

      Distance from launch site to closest coastline: 0.5097439631188213 km
```



Section 4

Build a Dashboard with Plotly Dash

Launch Success Count for all sites (in a pie chart)

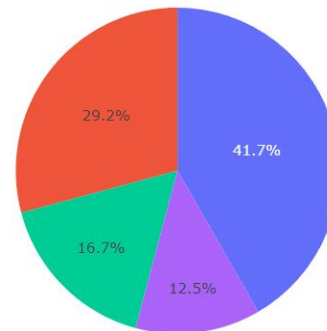
Key Findings:

- **CCAFS LC-40**: 29.2%
- **CCAFS SLC-40**: 12.5%
- **VAFB SLC-4E**: 16.7%
- **KSC LC-39A**: 41.7%
- The **KSC LC-39A** launch site has the highest number of successful launches, making up 41.7% of the total successes. This indicates that KSC LC-39A is a highly reliable site for SpaceX launches.

SpaceX Launch Records Dashboard

All Sites

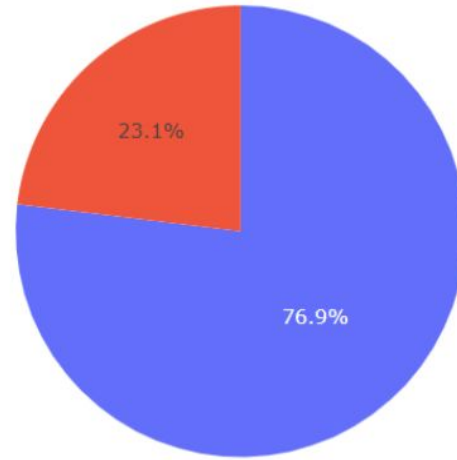
Total Success Launches by Site



■ KSC LC-39A
■ CCAFS LC-40
■ VAFB SLC-4E
■ CCAFS SLC-40

Payload range (Kg):





Pie chart for
the launch site
with highest
launch success
ratio

Key Findings:

- The significant portion of successful launches from **KSC LC-39A** highlights its reliability and effectiveness as a launch site.
- For **KSC LC-39A**:
 - **Class 1** (Successful Launches): 76.9%
 - **Class 0** (Unsuccessful Launches): 23.1%
- The high success rate (76.9%) for **Class 1** launches underscores the effectiveness and reliability of the KSC LC-39A site.

Key Insights from SpaceX Launch Data Dashboard

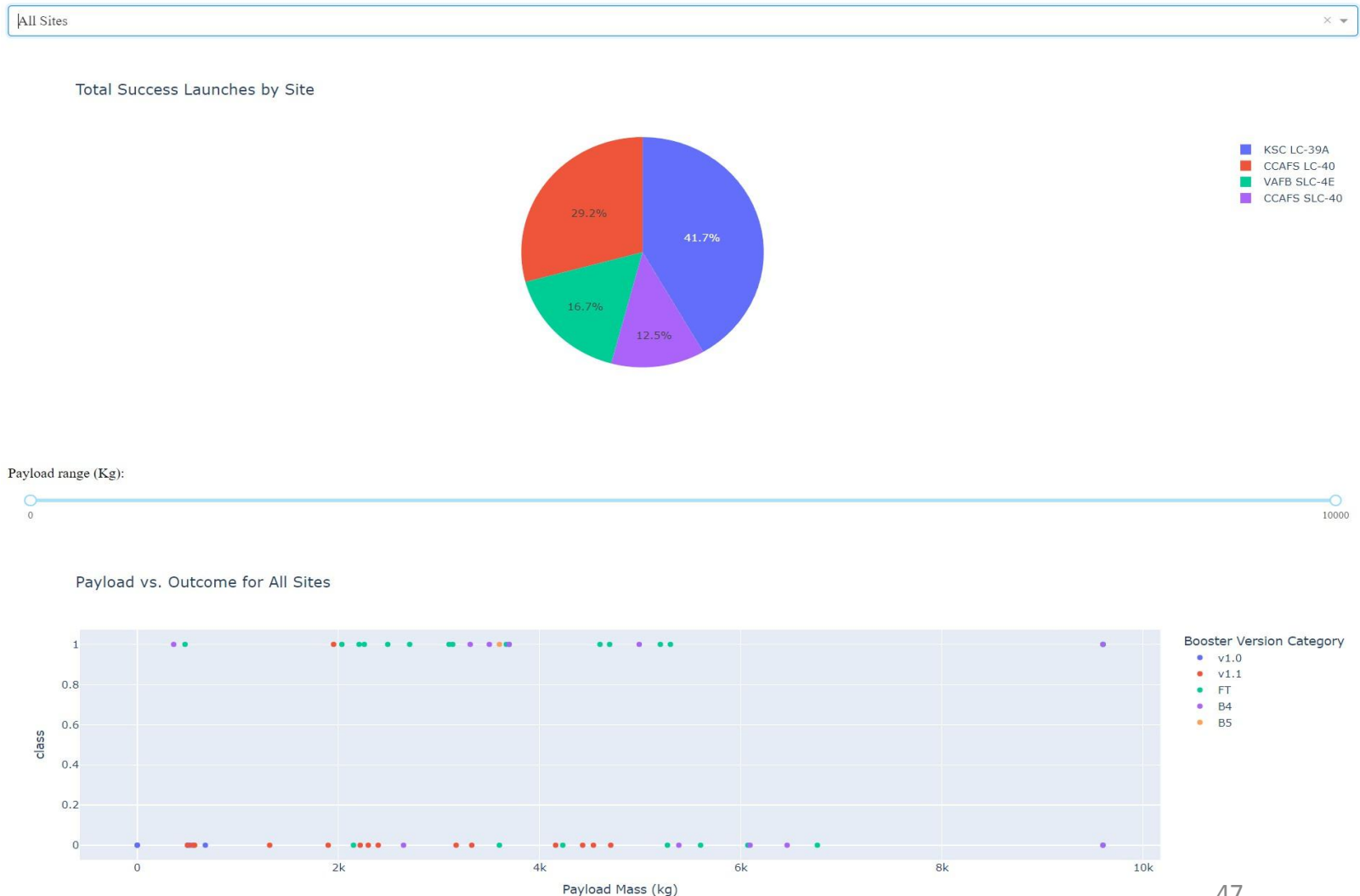
Launch Site Success Rates:

- **CCAFS LC-40** has the highest success rate with 43.7% of successful launches.
- This suggests that **CCAFS LC-40** is the most reliable launch site among the ones analyzed.
- Other sites like **KSC LC-39A**, **VAFB SLC-4E**, and **CCAFS SLC-40** have lower success rates, indicating variability in launch success across different sites.

Booster Version Performance:

- **Booster version "FT"** appears to be the most frequently used and has a high success rate across various payload masses.
- **Booster version "v1.0"** has fewer launches and may require further analysis to understand its performance.
- Overall, booster versions do not show a clear trend that higher payload masses correlate with lower success rates.

SpaceX Launch Records Dashboard



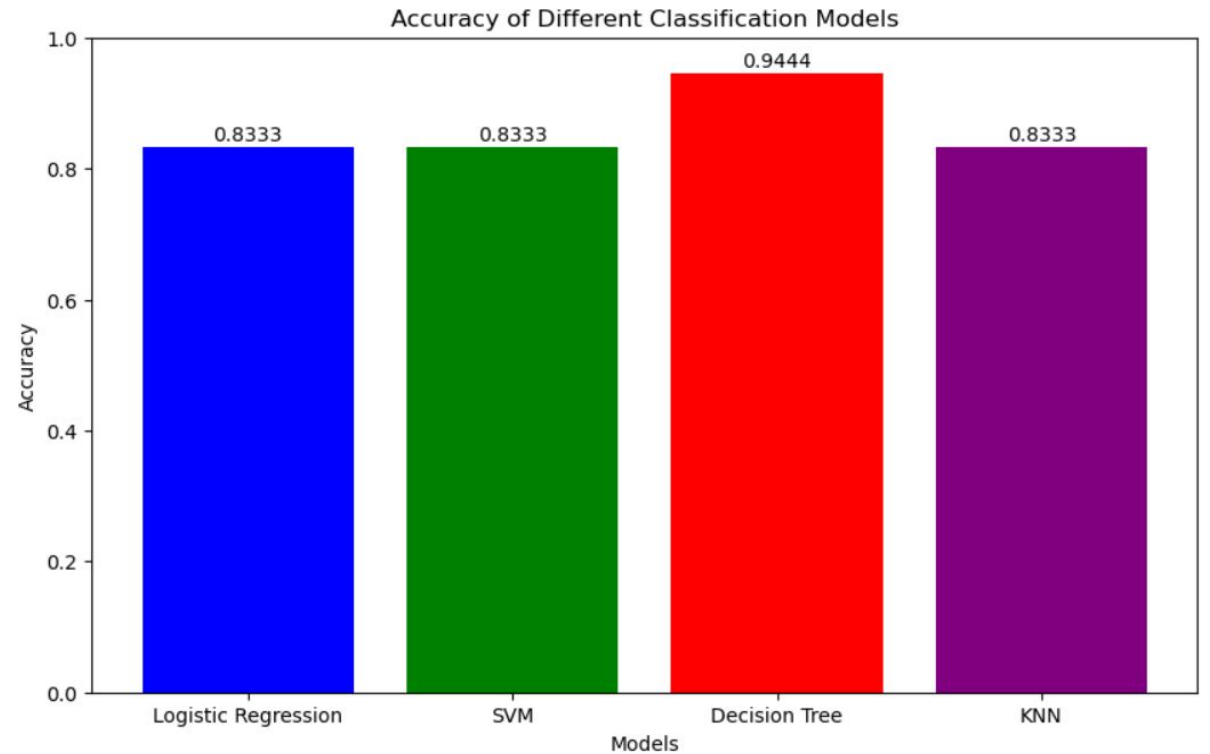


Section 5

Predictive Analysis (Classification)

Classification Accuracy

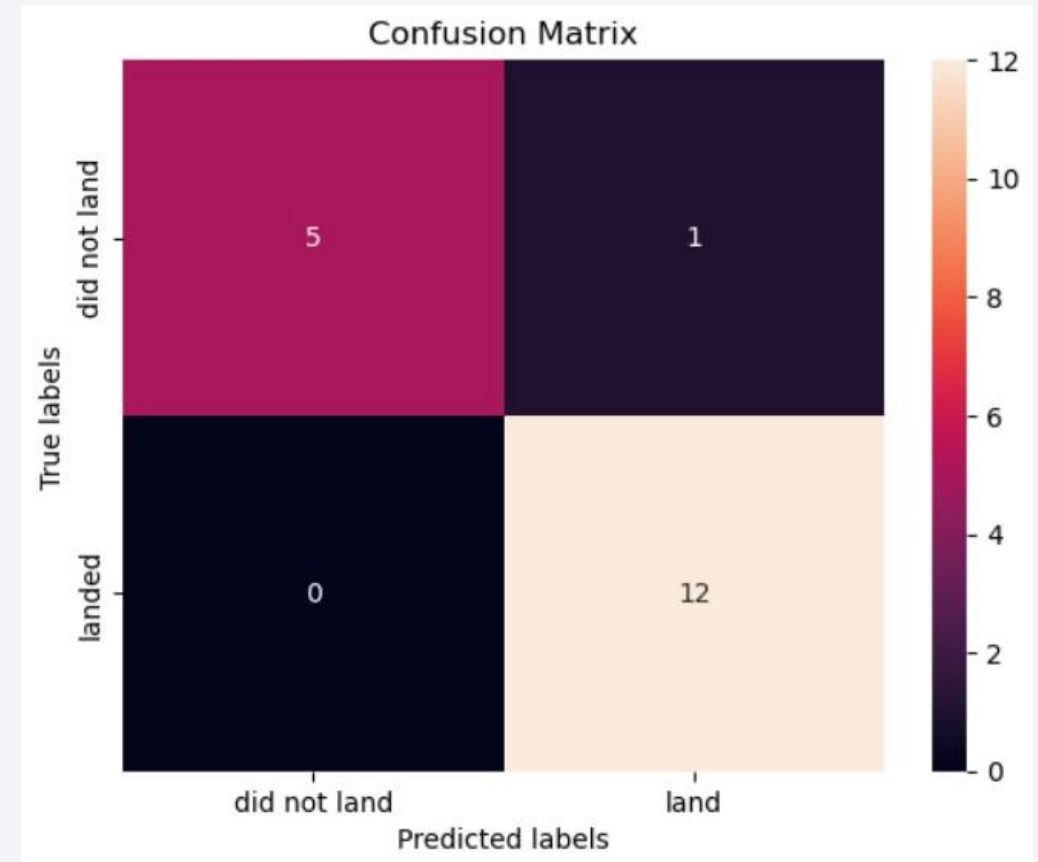
- Based on the results, the Decision Tree model has the highest classification accuracy on the test data, achieving an accuracy of 0.9444. This suggests that the Decision Tree model is better suited for this dataset compared to Logistic Regression, Support Vector Machine, and K Nearest Neighbors, all of which achieved an accuracy of 0.8333.



Confusion Matrix

Model Performance Analysis and Insights

- **Robust Accuracy:** The model achieves 94.44% accuracy, demonstrating strong proficiency in classifying Falcon 9 first-stage landings.
- **Safety Reliability:** Zero false negatives ensure consistent detection of successful landings, a critical requirement for aerospace operational readiness.
- **Strategic Error Management:** A single false positive is acceptable; in aerospace, the risks of false negatives (under-preparation) far outweigh the manageable costs of false positives (over-preparation).
- **Operational Alignment:** The model is effectively calibrated for industry needs, balancing performance with a slight optimization toward confirming successful landings for mission planning and cost estimation.



Conclusions

- **Launch Site Performance:** The "CCAFS LC-40" facility demonstrated the highest performance among all locations, responsible for 43.7% of successful missions. This performance disparity suggests that the site may benefit from superior operational procedures or more favorable environmental conditions.
- **Booster Reliability:** Analysis of the "FT" booster version indicates consistent success across a wide range of payload masses. This performance profile highlights the version's robustness and reliability, suggesting it is a strong candidate for future missions aiming to maximize success rates.
- **Influence of Payload Mass:** Our findings indicate no significant correlation between payload mass and launch failure. This implies that mission success is driven primarily by variables such as booster architecture and launch site infrastructure rather than the weight of the payload itself.

Conclusions

The integration of Folium and Plotly Dash for interactive data visualization offered critical insights into the operational and geographical trends surrounding SpaceX launches. By leveraging these visual analytics tools, we gained a more granular understanding of launch data, which supports data-driven decision-making for stakeholders.

Ultimately, combining predictive modeling with these interactive visualizations has identified essential variables impacting mission success. This approach establishes a strong framework for future aerospace industry assessments, helping to refine launch strategies and further the advancement of reusable rocket technology.

Thank you!

